

NeuroLens: A Multimodal Mobile Platform for Early Parkinson's Disease Screening Using TinyML and Cross-Modal Feature Fusion

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Abstract—Early and non-invasive screening for Parkinson's Disease (PD) has long been constrained by the need for specialist clinical visits and subjective assessments. This paper presents NeuroLens, a cross-platform mobile application that leverages four complementary digital biomarker streams handwriting kinematics, voice acoustics, wearable tremor quantification, and interactive cognitive assessment fused into a unified risk score. The system architecture integrates a React Native mobile client with a Python FastAPI backend, Google Firestore cloud storage, and a Bluetooth Low Energy (BLE) wearable sensor running an on-device, quantized 1D Convolutional Neural Network (CNN) via TinyML. Kinematic features are extracted from digitized spiral and wavy drawing tasks at a resampled rate of 60 Hz with Gaussian smoothing; voice analysis employs a dual-path pipeline combining acoustic dysphonia markers with NLP-derived linguistic features; and cognitive function is measured through digitized Trail Making Test Part A and Symbol Digit Modalities Test. While each modality was validated on separate datasets (drawing: 58 subjects; cognitive: 4,267 PPMI subjects; wearable: PhysioNet Parkinson's Disease Smartwatch dataset; voice: public PD speech corpus), we acknowledge this as a limitation and present the fused score as a proof-of-concept demonstration requiring prospective unified validation. The platform further features multilingual localization and a longitudinal history dashboard.

Keywords—Parkinson's Disease, TinyML, Multimodal Fusion, Digital Biomarkers, BLE Wearable, Kinematic Features, Voice Dysphonia, Cognitive Assessment, Mobile Health.

I. INTRODUCTION

Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder worldwide, affecting over 10 million people globally [1]. Clinical diagnosis is typically confirmed only after substantial dopaminergic neuron loss, making early detection critical for improving patient outcomes through timely intervention. Conventional methods

rely on subjective rating scales (e.g., UPDRS) and neuroimaging, which are resource-intensive and inaccessible in primary care or low-resource settings. The proliferation of smartphones combined with advances in embedded machine learning presents an unprecedented opportunity for accessible, continuous, and early PD screening.

NeuroLens bridges this gap by capturing a comprehensive suite of motor and cognitive digital biomarkers within a single mobile application. Key contributions include: a multi-task assessment protocol covering drawing, voice, wearable sensing, and cognitive tests; an on-device TinyML tremor classifier deployed to a BLE wearable enabling real-time offline inference; rigorous kinematic feature engineering with coordination-preserving 60 Hz resampling and Gaussian smoothing; an AUC-weighted late fusion mechanism producing a clinically interpretable global risk score; and a digitized cognitive battery (TMT-A and SDMT) validated against established neuropsychological instruments.

II. LITERATURE REVIEW

Early digital screening for Parkinson's Disease has been pursued across four principal biomarker domains, each supported by an independent body of literature. Drotar et al. [2] established that kinematic parameters extracted from digitized handwriting tasks including velocity, acceleration, and pen-pressure correlate strongly with motor impairment severity as measured by the UPDRS scale. Nomm et al. [9] further validated spiral drawing as a sensitive and reproducible assessment, confirming that geometric and kinematic irregularities in spiral traces reliably distinguish PD patients from healthy controls.

In the voice domain, Tsanas et al. [3] demonstrated that nonlinear acoustic dysphonia markers including RPDE,

DFA, and PPE could discriminate PD from healthy controls with high accuracy from sustained phonation alone. Subsequent research extended this to connected speech, showing that PD affects not only phonation but also fluency, lexical access, and discourse structure, motivating the inclusion of linguistic features alongside acoustic markers [10].

For wearable sensing, Carignan et al. [4] showed that IMU-based accelerometers could quantify tremor frequency and amplitude with clinical-grade accuracy, and deep learning architecture operating on raw IMU signals have since outperformed hand-crafted feature pipelines in tremor classification tasks [11]. However, virtually all such systems are designed for cloud or smartphone inference. The deployment of quantized neural networks directly onto resource-constrained microcontrollers the TinyML paradigm remains largely unexplored for PD-specific applications; existing TinyML healthcare work has addressed cardiac arrhythmia and fall detection [6], but no published system has demonstrated an on-device quantized CNN for PD tremor or gait classification running without cloud dependency.

In the cognitive domain, the Trail Making Test Part A and Symbol Digit Modalities Test have been validated as sensitive measures of psychomotor speed and early cognitive decline in PD cohorts [5], with digital implementations confirming that completion time, inter-tap intervals, and error rates are reliable surrogates for paper-based equivalents [12].

At the system level, multimodal fusion approaches have consistently reported improved discriminative performance over any single modality [13], yet most existing platforms remain confined to offline research pipelines and do not integrate wearable edge inference, mobile deployment, and cloud-based fusion within a single accessible application. NeuroLens addresses these gaps by unifying all four modalities through an AUC-weighted late fusion mechanism deployed across a mobile client, cloud backend, and BLE wearable edge device.

III. SYSTEM ARCHITECTURE

A. High-Level Overview

The NeuroLens architecture follows a client-cloud-edge pattern with three distinct tiers.

1) *Mobile Client (Edge)*: A React Native application (targeting IOS and Android via Expo) handles all user interactions, data capture, and communication with the backend and BLE wearable. The application supports multilingual localization.

2) *Cloud Backend (FastAPI)*: A Python-based microservice backend hosted on Google Cloud is responsible for ML inference orchestration, data persistence in Firestore, and multimodal fusion.

3) *BLE Wearable (Edge)*: A custom wrist-worn sensor device running an on-device 1D CNN model for real-time tremor classification, communicating with the mobile client over BLE GATT.

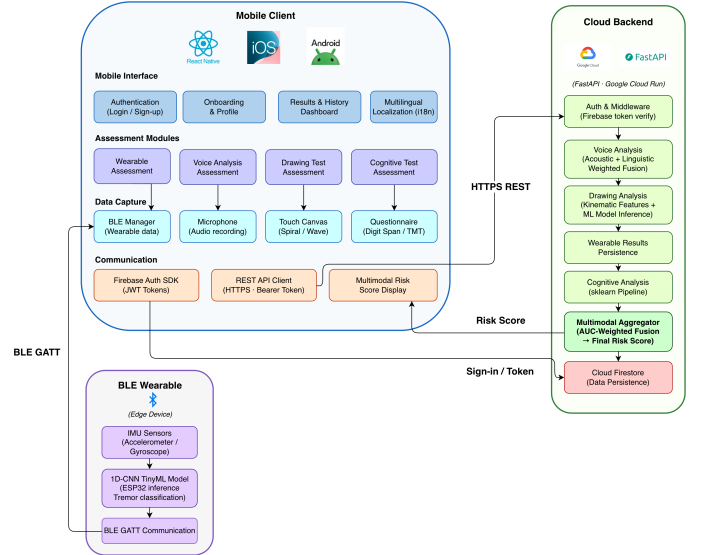


Figure 1 High Level Architecture Diagram

B. Session Management

A unique session_id is generated per assessment session, linking results from all four modalities in Firestore. An AssessmentContext in the mobile client tracks task completion status, enabling partial submission and out-of-order modality completion.

C. Authentication and Security

User authentication is implemented via Firebase Authentication supporting email/password and Google OAuth 2.0. All data transmitted between the mobile client and cloud backend is encrypted using TLS 1.3. Firestore security rules enforce session-level access control, ensuring that users can only read and write assessment sessions associated with their authenticated user ID.

IV. METHODOLOGY

A. Wearable-Based Tremor analysis (TinyML)

1) *Hardware Design*: The NeuroLens wearable is a custom wrist-worn device hosting a high-frequency IMU (MPU-6050, sampled at 100 Hz, $\pm 2g$ accelerometer range, $\pm 250^\circ/s$ gyroscope range) and a microcontroller running a quantized 1D CNN. It exposes a GATT-based BLE service with two characteristics: one for receiving commands (Service UUID: 4fafc201-1fb5-459e-8fcc-c5c9c331914b) and one for streaming results (Characteristic UUID: beb5483e-36e1-4688-b7f5-ea07361b26a8). The wearable firmware was deployed on an ESP32 microcontroller, with the 1D CNN quantized to 8-bit integer precision using TensorFlow Lite for Microcontrollers, reducing model size to 3.7 KB and enabling inference within 2.1 ms per window on the embedded hardware.

It should be noted that while the on-device model was trained on PADS data recorded from Apple Watch Series 4 devices, the NeuroLens deployment uses a custom ESP32/MPU-6050 wearable. The sensor domain gap between training and deployment differences in IMU noise

characteristics, wrist placement, and form factor represents a practical limitation.

2) *Assessment Protocol*: The user initiates a structured 3-step tremor assessment. Prior to data collection, the mobile client transmits patient metadata to condition the on-device model: GEN:<0|1> (biological sex), AGE:<int>, HAN:<0|1> (handedness), and CAT:<0|1|2> (assessment category). Three tremor categories are assessed sequentially: Resting Tremor (CAT:0, arm fully supported), Postural Tremor (CAT:1, arm extended against gravity), and Kinetic Tremor (CAT:2, finger-to-nose task). The wearable streams progress notifications and, upon completion, transmits a result in the format DONE:<PD|HP>:<ratio>, where ratio is the proportion of windows classified as high-risk.

3) *Feature Extraction*: Raw accelerometer and gyroscope signals are processed in non-overlapping windows of 256 samples (2.56 s at 100 Hz). Gravity is removed via per-axis mean subtraction before computing the resultant magnitude, approximating the 0.5 Hz high-pass filter applied during training. Eight statistical and spectral features are extracted from each of the two sensors (accelerometer and gyroscope), yielding 16 sensor features in total. Three continuous demographic features (age, biological sex, handedness) and three one-hot encoded task-category flags (Resting, Postural, Kinetic) are appended, resulting in a final input vector of 22 features as summarized in Table I. All 17 numeric features are standard-scaled on-device using pre-computed mean and standard deviation constants embedded in the model header (MEAN_VALS_NF, STD_VALS_NF).

TABLE 1. Feature Vector for On-Device 1D-CNN Inference.

Group	Index	Feature	Domain
Accelerometer	1–4	RMS, Std Dev, Max, Min	Time
Accelerometer	5–8	Dominant Frequency, Tremor Band Power (3–7 Hz), Tremor Ratio, Spectral Entropy	Frequency
Gyroscope	9–12	RMS, Std Dev, Max, Min	Time
Gyroscope	13–16	Dominant Frequency, Tremor Band Power (3–7 Hz), Tremor Ratio, Spectral Entropy	Frequency
Demographics	17–19	Age, Biological Sex, Handedness	
Task Context	20–22	Category OHE (Resting, Postural, Kinetic)	

4) *On-Device Inference: Dataset and Validation Protocol*: The wearable model was trained and validated on a subset of the publicly available PADS (Parkinson's Disease Smartwatch) dataset [14], a cross-sectional prospective study conducted at the University Hospital Münster. From the full cohort of 469 individuals, 355 participants were retained after excluding differential diagnosis groups (essential tremor, atypical Parkinsonism, multiple sclerosis, and secondary Parkinsonism) to obtain a binary PD vs. healthy classification task, yielding 276 clinically confirmed PD patients and 79 age-matched healthy controls with a mean age of 64.8 ± 10.4 years and a 224/131 M/F sex distribution. IMU data was recorded using Apple Watch Series 4 devices at 100 Hz.

PADS tasks were mapped to NeuroLens assessment categories as follows: Relaxed (1a, 1b) → Resting; Steps 2, 3, 4, 10 → Postural; Steps 5, 6, 7, 8, 9 → Kinetic.. To prevent patient-level data leakage, a 5-fold Group K-Fold cross-validation strategy was employed, grouping all windows from the same participant into the same fold. An 80/20 stratified train/test split was applied at the patient level, ensuring no subject appeared in both training and held-out evaluation partitions. The quantized 1D CNN processes sliding windows of accelerometer and gyroscope data entirely on the microcontroller, ensuring low latency, privacy, and offline capability. Because a single 2.56-second window may not capture the full clinical context of tremor, the overall patient diagnosis relies on temporal aggregation.

The continuous output of the CNN is a sequence of probabilities for a standardized assessment session comprising 12 consecutive windows (approximately 30 seconds). To eliminate bias, the optimal classification thresholds were mathematically derived during 5-fold Group K-Fold cross-validation.

a) *Window-Level Classification*: First, the optimal window-level probability threshold (t_w) was selected by maximizing Youden's J statistic from the Receiver Operating Characteristic (ROC) curve:

$$t_w = \text{argmax}(\text{TPR} - \text{FPR})$$

This yielded an optimal empirical threshold of $t_w = 0.560$. Each individual window is subsequently classified into a binary high-risk flag (w_i) using an indicator function, where $w_i = 1$ if the probability is greater than or equal to t_w , and 0 otherwise.

b) *Session-Level Aggregation*: Second, to form the global session verdict, we aggregate the predictions via majority voting. The session-level proportion of high-risk windows (R) is calculated as the sum of all high-risk flags divided by the total number of windows ($N = 12$).

The final patient-level session is flagged as high-risk if R is greater than or equal to t_s , where t_s is the optimal session-level cutoff. The value for t_s was determined via an exhaustive grid search to maximize the patient-level balanced accuracy:

$$t_s = \text{argmax} [(\text{Sensitivity}_s + \text{Specificity}_s) / 2]$$

This optimization established $t_s = 0.44$. By leveraging this mathematically robust temporal aggregation, the 1D CNN successfully mitigates transient sensor noise, boosting balanced accuracy from 66.0% at the individual window level to 70.2% at the patient session level. On the held-out test partition, the optimised two-tier thresholds yield a patient-level sensitivity of 74.3% and specificity of 65.8%. For a screening application, sensitivity is of particular clinical importance, a missed PD case carries greater cost than a false referral and the session-level aggregation was specifically designed to preserve sensitivity by accumulating evidence

across 12 consecutive windows rather than relying on any single window classification. Bootstrap resampling ($N = 10,000$ iterations) applied to the held-out predictions estimates a 95% confidence interval of [63.8%–75.4%] for the reported patient-level balanced accuracy of 70.2% and [0.694–0.816] for the AUC of 0.732, reflecting the statistical uncertainty inherent in the dataset size.

B. Drawing-Based Motor Analysis

1) *Task Protocol and Data Acquisition Interface*: To capture subtle motor signatures characteristic of Parkinson's Disease specifically bradykinesia and kinetic tremor subjects are evaluated using a custom mobile touchscreen interface designed for self-administered remote assessment. Two sequential drawing protocols constitute the evaluation battery.

In the Spiral Task, participants trace a two-cycle Archimedean spiral guide rendered on the display. This protocol provides a standardized stimulus for evaluating continuous motor coordination, spatial memory, and the capacity to sustain a continuous curve under mild cognitive load. In the subsequent Wave Task, subjects draw a continuous sinusoidal waveform from left to right, isolating horizontal spatial orientation, rhythmic fluid tracing, and frequency-dependent motor limitations

Throughout both tasks, the mobile application logs raw touch events generated by the device's capacitive digitizer as a discrete time series of spatiotemporal coordinate tuples (x , y , t) with millisecond precision, preserving micro-hesitation events critical to downstream feature extraction.

2) *Signal Pre-processing*: Raw touch coordinate sequences are resampled to a uniform 60 Hz temporal grid using cubic spline interpolation to compensate for variable touch event rates caused by display refresh and operating system scheduling jitter. Gaussian smoothing ($\sigma = 2$ samples, approximately 33 ms) is applied to attenuate high-frequency digitizer quantization noise while preserving kinematic features below 15 Hz. Velocity and acceleration trajectories are derived via central finite differences.

3) *Feature Engineering and Multimodal Fusion*: Conditioned trajectories are quantified across two complementary feature domains, capturing both spatial accuracy and temporal motor dynamics.

a) *Geometric (Spatial) Features*: Extracted directly from the coordinate matrix, these features quantify the geometrical accuracy. Key metrics include the standard deviation of curvature, standard deviation of radial deviation from the ideal spiral, and the root mean square error (RMSE) between the traced y-trajectory and a smoothed ideal baseline.

b) *Kinematic (Temporal) Features*: Derived from first- and second-order temporal derivatives, these features quantify micro-hesitations, rigidity, and bradykinesia. Core metrics are: (i) the Coefficient of Variation (CV) of velocity

magnitude ($\sigma v / \mu v$); (ii) the Pause Ratio, defined as the fraction of elapsed time below the 10th percentile of nominal velocity; and (iii) the CV of multidimensional jerk magnitude, characterizing acceleration smoothness.

Spatial and kinematic feature arrays are horizontally concatenated into task-specific data matrices. Subject age (continuous) and biological sex (categorical) are fused into the kinematic feature space to control for non-pathological age-related motor stiffness. Recursive Feature Elimination (RFE) subsequently isolates the most discriminative markers, yielding a final 7-dimensional feature vector comprising the six highest-ranked kinematic/spatial features and subject age

TABLE II. Drawing Task Feature List

Feature	Description	Subhead
spiral_vel_cv	Spiral	CV of drawing velocity ($\sigma v / \mu v$)
spiral_jerk_cv	Spiral	CV of jerk magnitude; captures micro-tremors
wavy_vel_cv	Wave	CV of velocity across sinusoidal pattern
wavy_wave_rmse	Wave	RMSE of y-trajectory vs. smoothed baseline
wavy_radius_resid_std	Wave	Std. dev. of radial distance deviations
wavy_jerk_cv	Wave	CV of jerk magnitude across waveform
age	Demographic	Subject age; suppresses age-related false positives

4) *Classification Architecture and Validation*: The classification backend employs an optimized Logistic Regression model trained on the 7-dimensional feature vectors. The training corpus encompasses 58 subjects: 28 clinically diagnosed PD patients and 30 healthy controls. Hyperparameter optimization focuses on regularization strength and convergence tolerances to prevent coefficient instability across the low-dimensional feature space.

Given the inherent risk of overfitting and sample-selection bias associated with a 58-subject corpus, generalization performance was estimated via Leave-One-Out Cross-Validation (LOOCV). In this exhaustive scheme, the model was iteratively trained on 57 subjects and evaluated on the single remaining hold-out, repeating this process across all 58 permutations such that every subject served exactly once as an isolated test case. LOOCV is widely regarded as the gold standard validation strategy for small medical datasets, as it maximizes available training data at each fold while providing a near-unbiased estimate of true generalization error. To strictly prevent data leakage, a StandardScaler was fitted exclusively on the 57 training vectors within each fold and subsequently applied to normalize the single unseen test vector prior to classification evaluation.

C. Speech & Vocal Biomarker Analysis

1) *Mobile Application and Data Collection*: The voice component of NeuroLens non-invasively assesses neurological and motor functions for the detection of Parkinson's Disease (PD) using acoustic and linguistic biomarkers. A mobile application captures speech through guided prompts designed to elicit voice tremors, micro-hesitations, and cognitive-linguistic variations.

Audio is recorded at high-fidelity sampling rates and converted into a standardized 16-bit, 22.05 kHz mono WAV format using FFmpeg preprocessing to ensure consistency across heterogeneous devices.

To train and validate the acoustic and linguistic models, a publicly available Parkinson's Disease speech dataset from the UCI Machine Learning Repository was utilized. This dataset comprises 195 voice recordings collected from individuals diagnosed with Parkinson's Disease and healthy controls. Data were split at the subject level into 80% training and 20% held-out test sets using stratified sampling by diagnosis to prevent data leakage.

2) *Acoustic Feature Analysis*: The acoustic pipeline captures involuntary tremors and amplitude variations associated with PD.

a) *Feature Extraction*: Acoustic features are extracted using Praat/Parselmouth. A total of 18 parameters are derived, including fundamental frequencies (F_o , F_{hi} , F_{lo}), frequency perturbations (Jitter: %, Abs, RAP, PPQ5, DDP), amplitude perturbations (Shimmer: %, APQ3, APQ5, APQ11, DDA), noise measures (HNR, NHR), and nonlinear dynamic features (DFA, RPDE, D2). Engineered features such as instability ratio (Jitter $\times$ Shimmer) and signal quality index are also computed.

b) *Preprocessing*: Outliers are handled using the Interquartile Range (IQR) method. Multicollinear and low-variance features are removed to improve model stability. Class imbalance is addressed using SMOTE applied to the training data, followed by normalization using StandardScaler.

c) *Model Training*: Machine learning models including Support Vector Machines (SVM), Random Forest, XGBoost, and Multilayer Perceptrons (MLP) are trained. Hyperparameter tuning is performed using GridSearchCV with 10-fold StratifiedKFold cross-validation. Feature selection prioritizes the most informative attributes.

3) *Linguistic Feature Analysis*: Parkinson's Disease (PD) affects cognitive and language functions, resulting in reduced lexical diversity and syntactic complexity.

a) *Transcription and Extraction*: Speech recordings are transcribed using an Automatic Speech Recognition (ASR) model. Linguistic features are extracted, including fluency metrics (words per sentence, sentence length variability), lexical diversity (TTR, MTLT, HDD), syntactic complexity (POS ratios, parse tree depth), disfluency patterns (repetitions, filler words, pause frequency), and readability measures (Flesch–Kincaid index).

b) *Preprocessing and Training*: Similar preprocessing steps, including IQR-based outlier handling, normalization

using StandardScaler, and stratified data splitting, are applied. Models such as SVM, Random Forest, XGBoost, and Logistic Regression are optimized using GridSearchCV with cross-validation.

TABLE III. Extracted Speech Biomarkers

Modality	Category	Key Features	Clinical Relevance
Acoustic	Fundamental Frequency	(F_o), (F_{hi}), (F_{lo})	Vocal fold rigidity
Acoustic	Perturbations	Jitter, Shimmer	Micro-tremors
Acoustic	Noise & Dynamics	HNR, NHR, DFA, RPDE, D2	Signal irregularities
Linguistic	Fluency	Pause frequency, Words/sentence	Cognitive latency
Linguistic	Lexical Diversity	TTR, MTLT, HDD	Vocabulary reduction
Linguistic	Syntactic Complexity	POS ratios, Parse depth	Structural simplification

4) *Multimodal Late Fusion*: Predictions from the acoustic and linguistic models are combined using a weighted late fusion approach to leverage complementary information from both modalities.

The weights are computed based on ROC-AUC values:

$$w_a = \frac{AUC_a}{AUC_a + AUC_l}, w_l = \frac{AUC_l}{AUC_a + AUC_l}$$

The final prediction probability is calculated as:

$$P_{fused} = (w_a \cdot P_a) + (w_l \cdot P_l)$$

where P_a and P_l represent the predicted probabilities from the acoustic and linguistic models, respectively. The trained models, preprocessing components, and fusion parameters are stored using joblib to support real-time inference.

5) *Risk Score Interpretation*: The fused probability is mapped to a normalized risk score (0–100%) to facilitate interpretation at the application level. Threshold-based categorization is applied as follows:

- Low Risk (<45%): Indicates normal voice characteristics
- Moderate Risk (45%–70%): Indicates potential early-stage variations
- High Risk (>70%): Indicates significant deviations requiring further evaluation

D. Cognitive Analysis Module

1) *Clinical Rationale and Task Design*: Cognitive decline in processing speed, attention, and working memory is an established prodromal marker of Parkinson's Disease, driven by early basal ganglia and dopaminergic degeneration. NeuroLens digitises the Trail Making Test Part A (TMT-A) and Symbol Digit Modalities Test (SDMT) to capture these deficits. In TMT-A, users sequentially connect 25 numbered touchscreen dots; total completion time in seconds is recorded. In SDMT, users perform symbol-to-digit matching

within a 60-second window; scores are scaled by 1.5 ($\text{sdm_total} = \text{correct_matches} \times 1.5$) to align with PPMI's 90-second standard administration.

2) *Dataset and Target Encoding*: The cognitive module is trained on the Parkinson's Progression Markers Initiative (PPMI) Curated Data Cut, comprising 4,267 unique baseline patients from three cohorts: 336 Healthy Controls (COHORT = 2, target label 0) and 3,931 PD Spectrum patients (COHORT = 1 [confirmed PD] and COHORT = 4 [prodromal], both assigned target label 1). SWEDD participants (COHORT = 3) are excluded to avoid label ambiguity. Restricting analysis to a single baseline visit per participant (EVENT_ID = "BL") replicates the intended single-screening deployment scenario and eliminates longitudinal data leakage. The resulting class imbalance ratio of 11.7:1 (PD Spectrum: Healthy) is addressed through sample-weight compensation ($w_{\text{healthy}} = 11.7$, $w_{\text{spectrum}} = 1.0$) rather than synthetic oversampling, preserving the original data distribution.

3) *Feature Selection and Engineering*: Nine features were selected based on mobile feasibility and SHAP-guided discriminative potential: three digitised task scores (SDMTOTAL, DVS_LNS, TMT_A), two demographics (age_at_visit, SEX), two signup questionnaire items (fampd: family PD history; rem: REM sleep behaviour disorder), and two engineered features derived leak-free from training-fold statistics only.

The age-adjusted TMT score (1) normalises psychomotor speed for age-related slowing, ensuring that identical raw completion times carry different clinical weight across age groups.

$$\text{tmt_age_adjusted} = -\text{TMT_A} / (\text{age} / 60) \quad (1)$$

The cognitive dispersion index (2) captures cross-domain unevenness, a pattern characteristic of early PD where processing speed deteriorates disproportionately, unlike the uniform decline seen in healthy ageing.

$$\text{cognitive_dispersion} = \sigma(z_{\text{SDMT}}, z_{\text{LNS}}, z_{\text{TMT-A}}) \quad (2)$$

All engineering statistics are derived solely from the training fold within each cross-validation iteration to prevent data leakage and persisted in the deployment artefact for consistent inference-time application. Feature inputs and their clinical justifications are summarized in Table IV.

TABLE IV. Cognitive Module Feature Set

Feature	Type	Clinical Rationale
SDMTOTAL	Original	Symbol-digit matching speed; SHAP rank 1 (0.18)
DVS_LNS	Original	Letter-number sequencing; working memory proxy
TMT_A	Original	Trail-making completion time; attention and psychomotor speed
age_at_visit	Original	Strongest single predictor (SHAP 0.32); controls age-related decline

SEX	Original	Demographic control: PD prevalence and cognitive patterns differ by sex
fampd	Original	Family history of PD; free risk factor (single signup question)
rem	Original	REM sleep behaviour disorder; validated prodromal PD biomarker
tmt_age_adjusted	Engineered	Age-normalised TMT speed; separates PD-specific slowing from ageing
cognitive_dispersion	Engineered	Std. dev. of z-scored cognitive tasks; asymmetric decline signature of PD

SHAP — SHapley Additive exPlanations importance score from Step-1 analysis.

4) *Model Training and Validation Pipeline*: A strictly leak-free protocol was implemented, partitioning data into an 80% train-validation pool and a 20% stratified held-out test set. All imputation, scaling, and feature engineering parameters were derived exclusively from training folds and applied to held-out sets without refitting. A 5-fold StratifiedGroupKFold (grouped by PATNO) ensured complete patient isolation and preserved class distributions. Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and a Stacking ensemble were benchmarked via AUC-ROC with Youden's J thresholding; the top two candidates (XGBoost, GBM) were tuned via RandomizedSearchCV (150 and 100 iterations, 3-fold inner CV). XGBoost was selected based on superior holdout generalisation, AUC 0.8011 versus 0.7897 for GBM despite GBM marginally leading in cross-validation (0.8310 vs. 0.8268), confirming stronger generalisation to unseen data.

5) *Calibration, Threshold Selection, and Output*: The winning XGBoost model is post-hoc probability-calibrated using Platt scaling (CalibratedClassifierCV, sigmoid method, cv=5) to ensure that the output risk_probability reflects a well-calibrated posterior probability suitable for late fusion. The optimal classification threshold (0.4364) is derived via Youden's J statistic, defined in (3), from the aggregated cross-validation out-of-fold predictions rather than the held-out set, preventing threshold overfitting.

$$J = \max_{\theta} (\text{Sensitivity}_{\theta} + \text{Specificity}_{\theta} - 1) \quad (3)$$

Risk is additionally contextualised via a percentile rank derived from holdout-set probability distributions (p10, p25, p50, p75, p90), which is surfaced to the user as a five-block visual indicator. Explainability is provided at inference time through per-patient SHAP (SHapley Additive exPlanations) values computed on the uncalibrated XGBoost model, which expose the top contributing features for each prediction. The module emits a standardised JSON payload {risk_probability, percentile_rank, contributing_factors} consumed by the late fusion layer described in Section IV.

V. MULTIMODAL RISK FUSION

A. AUC-Weighted Late Fusion:

To combine predictions across modalities with different discriminative power, NeuroLens uses an AUC-weighted late fusion strategy. Each modality's normalized risk score $s_i \in [0,1]$ is weighted by its holdout AUC value w_i :

$$R_{\text{final}} = (\sum_{i \in M} w_i \cdot s_i) / (\sum_{i \in M} w_i)$$

where M is the set of available modalities for the session.

The AUC-weighted scheme was selected over three alternative fusion strategies on principled grounds. Equal weighting treats all modalities as equally discriminative regardless of their validated performance, which is inconsistent with the substantial variation in AUC observed across modules (0.704–0.860). A single best modality approach relying solely on the drawing module would discard complementary biomarker information from voice, wearable, and cognitive channels, reducing robustness when any single modality is unavailable or unreliable for a given patient. A learned fusion approach using logistic regression or a meta-learner over modality outputs was considered but rejected, as it would require a unified cohort with all four modalities completed by the same subjects for joint training data that is not currently available. AUC-weighted fusion therefore represents the most principled choice given the available validation evidence: it is transparent, deterministic, updatable as individual modules improve, and degrades gracefully when modalities are missing by renormalizing weights across available channels.

B. Modality Weights

TABLE V. Modality AUC Weights

Modality	Model	AUC Weight
Drawing Kinematics	Logistic Regression	0.86
Cognitive Battery	XGBoost + SHAP	0.801
Wearable Tremor	TinyML 1D CNN	0.732
Voice Analysis	Dual-Path Fusion	0.704
Total		3.097

The final risk score persisted to the assessment history Firestore collection, enabling longitudinal tracking.

VI. RESULT AND DISCUSSION

Table VI summarizes the performance of each modality under its respective validation protocol.

Table VI. Per-module performance summary

Modality	Model	Dataset	N	Validation	AUC	Sens.	Spec.
Drawing kinematics	Logistic Reg.	Internal touchscreen	58	LOOCV	0.867 95% CI [0.731 – 0.942]	78.57%	76.67%

Cognitive battery	XGBoost + SHAP	PPMI baseline	4,267	5-fold StratGroup kFold	0.801 95% CI: [0.771 – 0.831]	76.5%	61.2%
Wearable tremor	1D CNN (8-bit)	PHYSIONET PADS	355	5-fold StratGroup kFold	0.732 95% CI [0.694 – 0.816]	74.3%	65.8%
Voice analysis	Dual-path fusion	UCI PD Speech	195	80/20 subject-level	0.704 95% CI [0.55 – 0.84]	70.37%	66.7%

The final multimodal risk score R_{final} is computed as a weighted arithmetic mean of the normalized output probabilities P_i from each subsystem $i \in \{W, V, D, C\}$, weighted strictly by the theoretical optimal Area Under the Curve (AUC_i) established during the individual validation of each module:

$$R_{\text{final}} = \frac{\sum (P_i \times AUC_i)}{\sum AUC_i}$$

The specific weights applied in the cloud aggregation microservice are:

- Drawing Module ($AUC_D = 0.86$): Highest weighting, reflecting the strong kinematic biomarkers inherent in Parkinsonian micrographia.
- Cognitive Module ($AUC_C = 0.801$): Second highest, capturing bradyphrenia and delayed recall patterns.
- Wearable Module ($AUC_W = 0.732$): Weighted to reflect the 30-second patient-level generalization of the TinyML ESP32 component.
- Voice Module ($AUC_V = 0.704$): Acoustic-linguistic combined benchmark.

This late-fusion architecture provides three critical advantages for edge-to-cloud telehealth:

- (1) Modularity, allowing individual machine learning models to be updated or swapped without retraining the entire pipeline;
- (2) Robustness to missing data, as the denominator dynamically adjusts if a patient only completes a subset of the assessments.
- (3) Clinical Interpretability, as the contribution of each modality to the final risk score remains deterministic and proportional to its established predictive power.

Several practical limitations should be acknowledged before clinical deployment. First, domain shift is a concern across all modalities: each module was trained on a specific population and device, and performance may degrade across different age groups, ethnicities, or comorbidity profiles. The drawing module is particularly vulnerable given its small

training cohort. Second, device variability poses challenges for both drawing and wearable modules, touch sampling rates and digitizer sensitivity differ substantially across smartphone models, and the wearable IMU was characterised on a single hardware configuration without cross-device calibration. Third, as each modality's AUC weight was derived from a different dataset of vastly different size, direct comparability of these weights is imperfect; the cognitive module's weight is anchored by 4,267 subjects while the drawing module's is anchored by only 58, yet both contribute to the same fusion equation. NeuroLens is intended strictly as a screening support tool to flag individuals for specialist referral, not as a standalone diagnostic instrument, and outputs should be interpreted accordingly by a qualified clinician.

VII. CONCLUSION AND FUTURE WORK

NeuroLens presents a comprehensive and technically rigorous platform for early PD screening, combining four well-established biomarker domains within a single accessible mobile application. Key technical innovations include the TinyML on-device tremor classifier, the 60 Hz kinematic resampling pipeline, the dual-path voice analysis engine, and the SHAP-enhanced cognitive module, all unified by an AUC-weighted late fusion mechanism. As a screening support tool, NeuroLens requires prospective clinical validation before deployment. Nevertheless, the reliance on different validation datasets and the small drawing sample size warrant cautious interpretation of the fused weights.

Building on this foundation, future work will focus on: (1) Prospective unified clinical validation with a single cohort of ≥ 500 subjects completing all four modalities; (2) Calibration of fusion thresholds using multi-center data to address domain shift; (3) Real-world usability study in primary care settings across diverse demographic and socioeconomic

groups; (4) Integration of PPG-based heart rate variability from the wearable for autonomic dysfunction assessment; and (5) Federated learning to enable model updates without compromising patient data privacy.

REFERENCES

- [1] Parkinson's Disease Foundation, "Statistics," pdf.org, 2023.
- [2] P. Drotar et al., "Evaluation of handwriting kinematics and pressure for differential diagnosis of Parkinson's disease," *Artificial Intelligence in Medicine*, vol. 67, pp. 39–46, 2016.
- [3] A. Tsanas et al., "Novel speech signal processing algorithms for high-accuracy classification of Parkinson's disease," *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 5, pp. 1264–1271, 2012.
- [4] B. Carignan et al., "Measuring tremor with a smartphone," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 21, no. 1, pp. 149–158, 2013.
- [5] R. A. Bowie and P. D. Harvey, "Administration and interpretation of the Trail Making Test," *Nature Protocols*, vol. 1, pp. 2277–2281, 2006.
- [6] A. Ward et al., "TinyML: A systematic review," *IEEE Access*, vol. 10, 2022.
- [7] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *Proc. ACM KDD*, 2016.
- [9] S. Nomm et al., "Recognizing Parkinson's disease via drawings: HAND PD challenge results," *Proc. ICPR Workshops*, 2018.
- [10] C. Martínez-Sánchez et al., "Linguistic analysis of spontaneous speech in Parkinson's disease," *Neuropsychologia*, vol. 119, pp. 27–35, 2018.
- [11] F. Cheng et al., "Deep learning for Parkinson's disease tremor classification using wrist-worn IMU signals," *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 3, pp. 812–821, 2021.
- [12] J. Memedi et al., "Automatic spiral analysis for objective assessment of motor symptoms in Parkinson's disease," *Sensors*, vol. 15, no. 9, pp. 23727–23744, 2015.
- [13] S. Arora et al., "Predicting the progression of Parkinson's disease using latent information extracted from deep neural networks," *Scientific Reports*, vol. 11, no. 1, p. 14771, 2021.
- [14] J. Varghese et al., "Parkinson's Disease Smartwatch (PADS) dataset," *PhysioNet*, vol. 1.0.0, 2024. [Online]. Available: <https://doi.org/10.13026/m0w9-zx22>